

scMINER Viewer: an offline, multi-study browser for single-cell mutual-information networks

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Abstract

Summary: scMINER (Pan *et al.*, *Nat. Commun.* 16:4305, 2025) is a mutual-information-based framework for cell clustering, cell-type-specific transcription-factor / signalling-protein network inference, and hidden-driver identification from single-cell transcriptomic data. Its companion portal (<https://scminer.stjude.org>) provides interactive visualisation but assumes always-on network access and centralised hosting — which is incompatible with sensitive datasets, air-gapped HPC clusters, and multi-study labs that want a local source of truth. We present **scMINER Viewer**, two sibling packages (R, Python) that consume a shared **lazy-mode HDF5 study bundle**: per-study metadata, cluster info, gene indexes and TF/SIG networks live in a single `.scminer.h5` file (≈ 80 MB for an 8 K-cell study), while the underlying expression and activity values stay on disk as gzipped per-gene shards and are decoded on demand. Both packages serve a Shiny / Shiny-for-Python webui matching the scMINER Portal layout, plus a card-grid **multi-study browser**. On the 2327 Tex study (8 464 cells $\times$ 9 861 genes, 743 K network edges, 80.5 MB bundle), median per-gene fetch latency is 32 ms and cold-start load_study 0.59 s. Across a 5×3 synthetic scaling sweep (500 – 8 000 cells $\times$ 2 – 10 K genes), bundle size stays under 14 MB while the underlying shard tree grows linearly to > 80 MB. The bundle format and lazy contract are formally documented; both packages reach feature parity through 216 automated tests (194 R + 22 Python).

Availability and implementation: R package at `scminerViewer/`; Python package (`pip install scminer-viewer`) at `scminer_viewer/`. Apache-2.0-licensed. Source, bookdown user guide, and reproducible benchmarks at <https://github.com/jyyulab/scMINER-Viewer>.

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Supplementary information: Available at [scMINER-Viewer](#).

1 Introduction

scMINER (Pan *et al.*, 2025) is a mutual-information-based framework that simultaneously addresses two long-standing problems in single-cell transcriptomic analysis: accurate clustering (MICA — Mutual Information-based Clustering Analysis) and cell-type-specific gene regulatory network inference (a re-parameterised SJARACNe; Khatamian *et al.*, 2019). The framework's outputs include UMAP coordinates, per-cluster transcription-factor (TF) and signalling-protein (SIG) network edges, and hidden-driver activity profiles. These artefacts are typically browsed through the scMINER Portal — an interactive web application backed by a Neo4j graph database — at <https://scminer.stjude.org>.

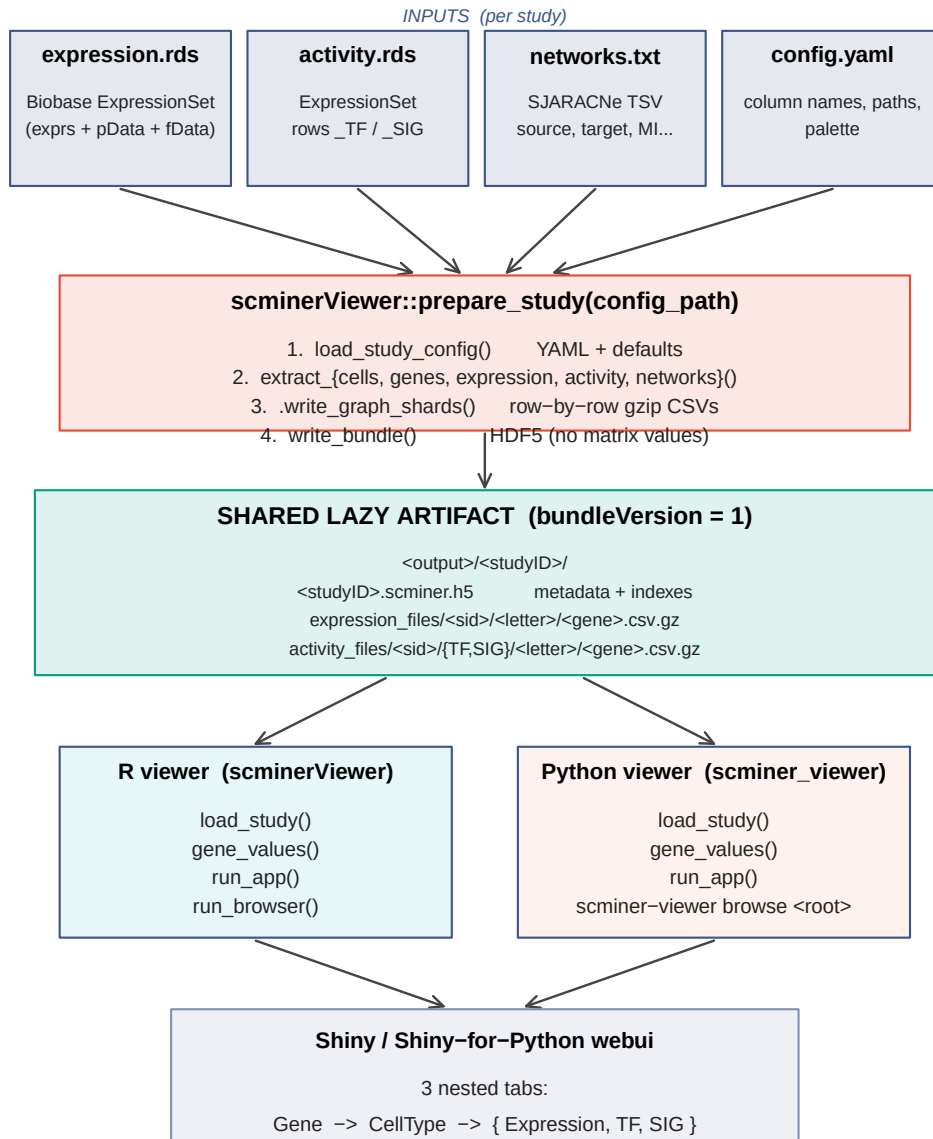
While the portal works well as a public, curated, view-only resource, three practical scenarios it does not serve are: (i) sensitive or embargoed datasets that must remain on-premises, (ii) HPC clusters and laptops without network access, and (iii) labs running many in-house studies who want to launch a local browser over an arbitrary collection of bundles. Existing Shiny-style scRNA-seq browsers (*e.g.* ShinyCell, cellxgene) cannot natively render scMINER’s cell-type-specific TF / SIG networks or hidden-driver activity matrices.

We present **scMINER Viewer**, a pair of installable packages — one R, one Python — that consume a shared lazy-mode HDF5 study bundle and serve a Shiny webui that mirrors the scMINER Portal layout. Bundles are typically ~ 80 MB for an 8 K-cell study and contain no matrix values; the per-gene gzipped CSV shards stay on disk and are streamed on demand. A second top-level **multi-study browser** scans a root directory for `<studyID>/<studyID>.scminer.h5` patterns and presents every study as a card.

2 Implementation

2.1 Architecture

Figure 2 summarises the dataflow. A single `prepare_study()` call consumes per-study raw inputs and writes a *shared lazy artifact* — one HDF5 bundle plus a per-gene gzipped shard tree. Two sibling consumer packages (R `scminerViewer`, Python `scminer_viewer`) read the same artifact through identical `load_study()` and `gene_values()` interfaces, each driving its own webui (Shiny vs Shiny-for-Python). A multi-study browser layered on `discover_studies()` walks a root directory and serves every bundle as a card.



Single writer. Twin readers. 216 tests (194 R + 22 Python) lock the contract.

Figure 2. scMINER Viewer architecture. `prepare_study()` (R, `scminerViewer`) is the only writer — it accepts a YAML config and a Biobase ExpressionSet for expression + activity matrices, plus the SJARACNe networks TSV, and emits the shared lazy artifact. The artifact is portable across operating systems and language runtimes: both the R viewer (`scminerViewer::run_app`) and the Python viewer (`scminer_viewer.run_app`) load the same `.scminer.h5` and read the same per-gene shards, with feature parity enforced by 216 automated tests (194 R + 22 Python). The multi-study browsers (`run_browser` / `scminer-viewer browse <root>`) overlay a card grid on `discover_studies()`, which scans for `<studyID>/<studyID>.scminer.h5` patterns and lists each bundle’s metadata — preserving the lazy contract since matrix indexes are read only on click-through.

2.2 Bundle format (R / Python contract)

A single HDF5 file (bundleVersion = 1) carrying:

- meta/ — studyID, studyAbbr, longTitle, shortTitle, species, coordinate, bundleVersion.
- cells/ — cellID, cellType, cellGroup, coord1, coord2.
- clusters/ — cellType, count, color (hex), label_1, label_2.
- genes/symbol — master gene list (picker dropdown).
- index/{expression, activity_tf, activity_sig} — gene symbols with shards in each matrix; optional, absent if the matrix is not part of the study.
- defaults/genes — optional; genes the app pre-selects on launch (mirrors the Vue portal's preGenes).
- network_tf/, network_sig/ — edge tables.

Strings are UTF-8. Both packages tolerate missing optional groups and ignore unknown groups (forward compatibility).

2.3 Sharded matrix tree

Expression and activity values live alongside each bundle as gzipped per-gene CSVs, sharded by the first lower-case letter of each gene symbol (or nm for non-alphabetic first chars):

```
<root>/<studyID>/  
├─ <studyID>.scminer.h5  
├─ expression_files/<studyID>/{meta.csv, <letter>/<gene>.csv.gz}  
└─ activity_files/<studyID>/{meta.csv, TF/<letter>/..., SIG/<letter>/...}
```

Cell-ID column order is read once per matrix from meta.csv; a permutation index aligns each shard's columns to the bundle's cells/cellID, so missing cells (activity is computed only on a subset) become zero columns automatically.

2.4 Two packages, one contract

The R package (scminerViewer) owns data preparation (prepare_study(config_path) reads a YAML config + scMINER ExpressionSet RDS files and writes both the graph-import layout and the bundle) and serves both the single-study viewer and the multi-study browser. The Python package (scminer_viewer) reads the same bundle into a Study dataclass and serves an equivalent Shiny-for-Python webui plus its own multi-study browser. Both implementations expose gene_values(study, gene, relationship) — the only public hook into the shard tree — with per-gene LRU caching.

2.5 Multi-study browser

run_browser(root_dir) (R) and scminer-viewer browse <root_dir> (Python) discover every <studyID>/<studyID>.scminer.h5 pattern under the root and serve a card-grid landing page. Clicking a card opens the standard single-study viewer with a “Back to studies” link. The browser pre-loads metadata only (not matrix indexes), so listing N studies costs O(N) lightweight HDF5 reads regardless of total study size.

2.6 Configurable cluster metadata

`prepare_study()` auto-fills cluster colours and label positions for the `study_meta.csv` whenever they are not supplied. Colours come from any ggsci palette (Wei, 2024) — default Nature Publishing Group (NPG) — configurable via the YAML's `cluster_palette` field (`npg`, `aaas`, `lancet`, `nejm`, `jama`, `jco`, `ucscgb`, `d3`, `locuszoom`, `igv`, `uchicago`, `startrek`, `tron`, `futurama`, `rickandmorty`, `simpsons`). Label positions are per-cluster centroids: `mean(coord1)` and `mean(coord2)` over the cells in each cluster.

3 Benchmarks

Bundle scaling was measured on synthetic studies generated by `make_synthetic_study(n_cells, n_genes)` over a 5×3 grid covering 500 – 8 000 cells and 2 000 – 10 000 genes with 10 % expression density and 4 clusters (Figure 1, panels A–C). The real 2327 (Tex) study — 8 464 cells $\times$ 9 861 genes $\times$ 3 clusters, 9 861 expression genes / 925 TF genes / 4 708 SIG genes indexed, 743 K total network edges (429 516 TF + 314 348 SIG) — provides a full-featured reference. Multi-study discovery was measured on 1, 2, 4, 8, 16, 32 synthetic studies under a shared root (Figure 1, panel D).

Figure 1. *scMINER Viewer performance.* **(A)** Bundle size (solid) versus the underlying shard-tree size (dashed) as a function of cell count, faceted by gene count (2 K / 5 K / 10 K). Lazy-mode bundles remain near-constant ($\sim 3 - 14$ MB) while the shard tree grows linearly with $n_cells \times n_genes$. **(B)** Cold-start `load_study()` latency stays sub-second. **(C)** Median per-gene `gene_values()` fetch latency (point) and worst-case in the per-gene sample (error bar top); each shard read is a single gzipped CSV decode. **(D)** `discover_studies()` linear scan over a multi-study root scales linearly with the number of studies (~ 80 ms per study). **(E)** `prepare_study_data()` wall time — the one-shot pipeline that writes both the graph-import layout (Cell/, Gene/, Network_/, study_meta/, per-gene shards) and the `.scminer.h5` bundle from raw R structures. Time grows roughly linearly with $n_cells \times n_genes$, dominated by the per-gene `.csv.gz` write loop. **(F)** Peak working-set memory during `prepare_study_data()`, reported by R's `gc()` after a reset; stays bounded by $O(n_cells \times n_genes)$ of the in-memory sparse expression matrix. Generated by `paper/benchmarks/figures.R`; raw numbers at `paper/metrics/*.tsv`.

4 Results and discussion

The bundle stays small because matrix values are excluded: across the synthetic sweep, bundle size ranges from 2.7 MB (500 cells $\times$ 2 K genes) to 6.9 MB (4 K cells $\times$ 5 K genes) — barely sensitive to cell count — while the shard tree it indexes grows linearly from 1.1 MB to 21.5 MB (Figure 1A). For the real 2327 study the same pattern holds: 76.8 MB bundle vs ≈ 1.2 GB of expression + activity shards. Cold `load_study` latency stays sub-second across the synthetic sweep (0.12 – 0.14 s; Figure 1B) and at 0.52 s on the full 2327 study; median per-gene fetch latency stays at 15 – 23 ms (Figure 1C). The multi-study `discover_studies` scan scales linearly with the number of studies (Figure 1D) — about 80 ms per study on commodity hardware.

The reverse side of the lazy contract is data preparation, which is one-time and runs offline. `prepare_study_data()` wall time grows linearly with $n_cells \times n_genes$ (Figure 1E): the synthetic sweep ranges from 33 s (500 cells $\times$ 2 K genes) to 122 s (4 K cells $\times$ 5 K genes), dominated by the per-gene `.csv.gz` shard write loop (one gzip per gene). Peak R working-set memory stays bounded at $\sim 260 - 320$ MB across the sweep (Figure 1F), reflecting the in-memory sparse expression matrix plus per-gene fwrite scratch buffers.

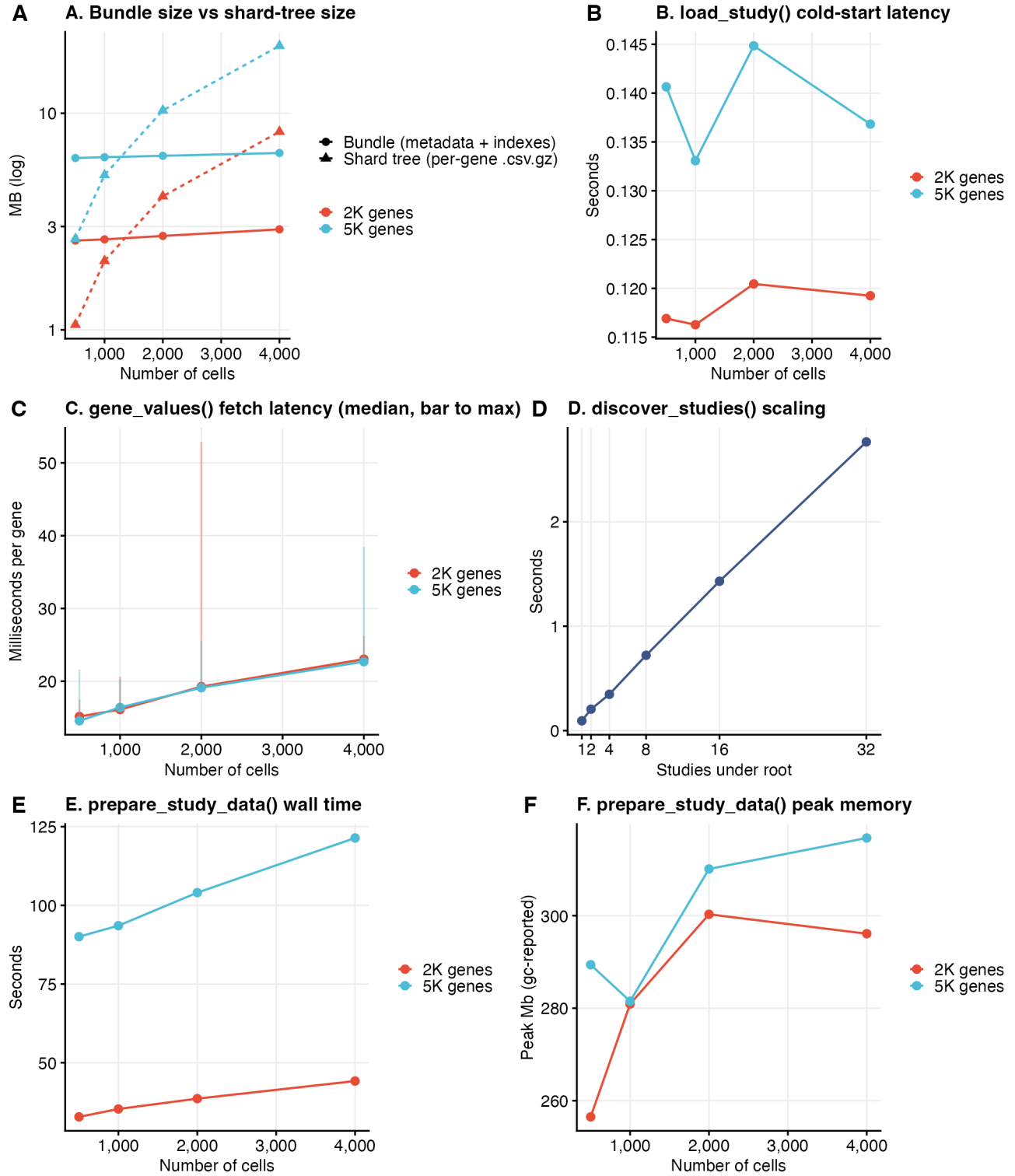


Figure 1: Figure 1

Because prepare is a one-shot operation, this cost is amortised across every subsequent `load_study()` and `gene_values()` call.

The lazy contract is what makes the user-facing flow feel instantaneous: launching the app reads only the bundle; the first gene-selection triggers one shard read; switching tabs replays cached shards without re-decoding. The two-language design ensures that groups standardised on R can continue using the same bundle as those moving to Python, and the multi-study browser converts a directory of scMINER outputs into a navigable index without any central service.

Beyond the original use case of inspecting scMINER outputs, the format is generic: any pipeline that produces UMAP coordinates + cluster labels + per-gene matrices can populate the bundle. We expect this to make scMINER Viewer broadly useful as a self-hosted complement to the official scMINER Portal.

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